AVISTA Model Report

An extensible desktop platform for tabular machine learning and deep learning analytics.

Project: mushroom

Target: poisonous

Original dataset rows: 8124

Features: 13

Original split rows: 5686 train / 813 validation / 1625 test

Effective training rows: 5686

Imbalance method: smote

Validation role: deep-model early stopping and checkpoint selection

Test role: held-out final evaluation after fitting and checkpoint selection

CV role: outer validation folds are used only for fold scoring

Cross-validation: 5 folds

Models trained: 5

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AVISTA version: 1.1.0

AVISTA release date: September 20, 2026

TabPFN 2.5 input: raw DataFrame with TabPFN-owned preprocessing

TabPFN external resampling: No

TabPFN available/effective training rows: 5686/5686

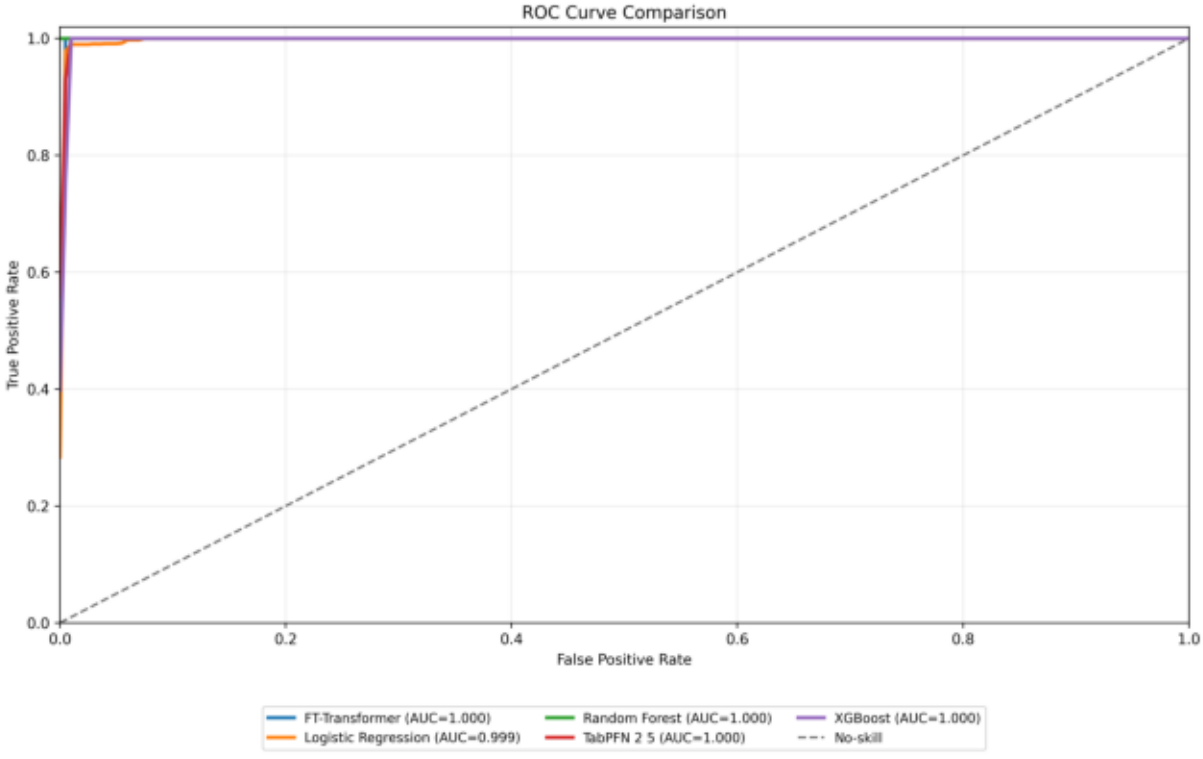
TabPFN supported row limit: 50000

Unless otherwise stated, model comparison figures and diagnostic tables are based on the test set.

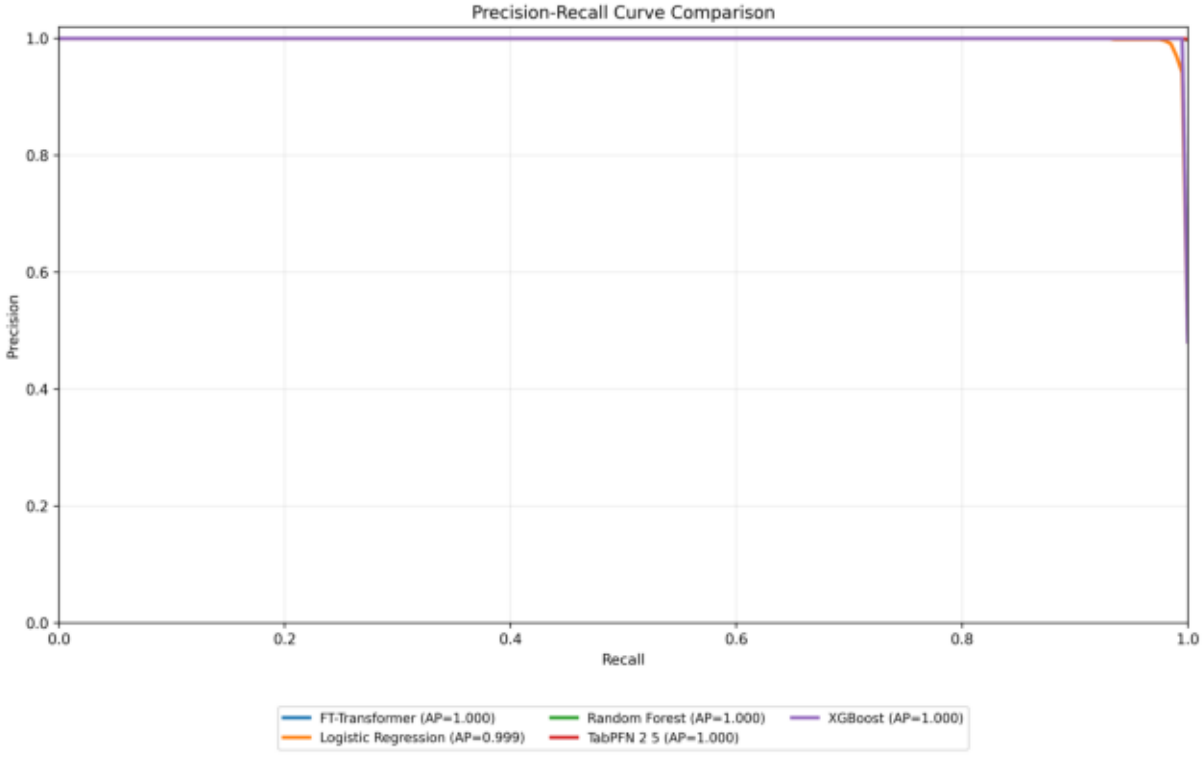
Model Performance Summary

Model	Status	Validation Accuracy	Validation Macro-F1	Test Accuracy	Test Macro-F1	ROC-AUC	Saved
Logistic Regression	trained	0.9865	0.9864	0.9760	0.9760	0.9993	Yes
Random Forest	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes
XGBoost	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes
FT-Transformer	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes
TabPFN 2.5	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes

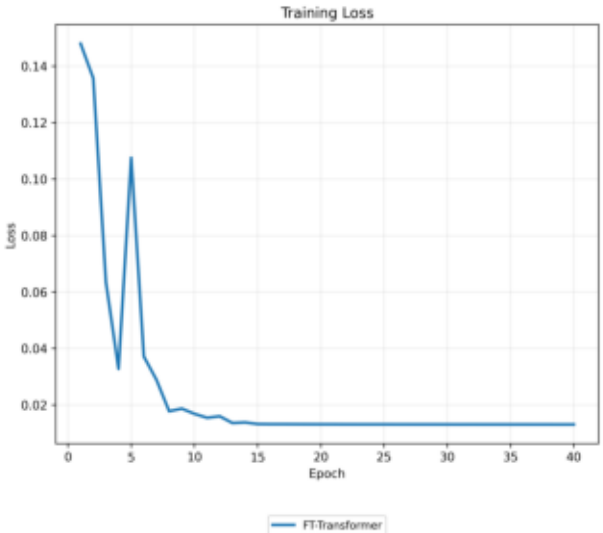
ROC Curve Comparison

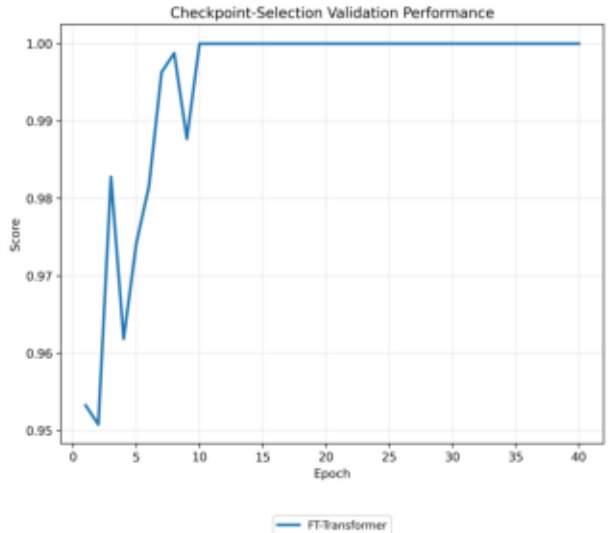


Precision-Recall Curve Comparison

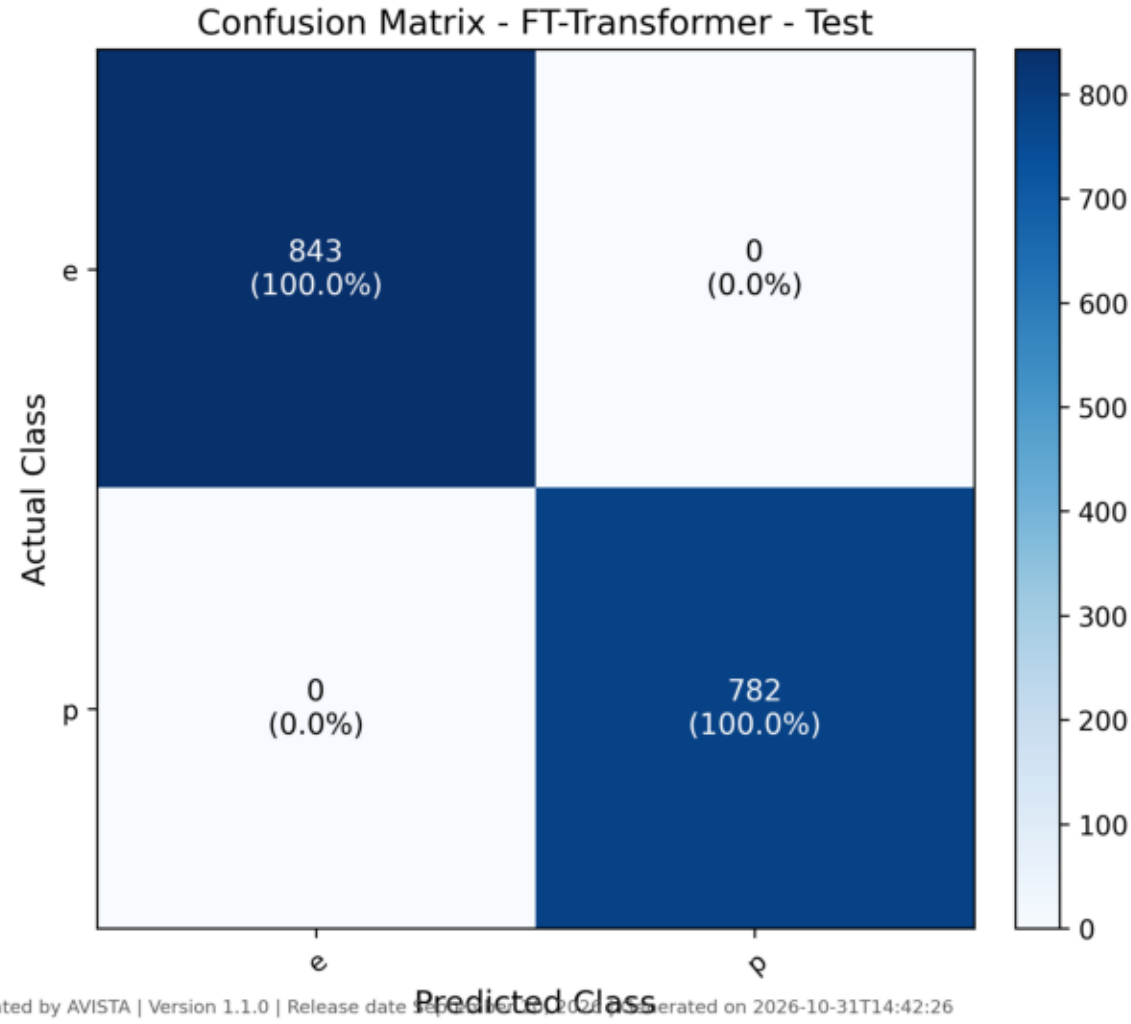


Deep Learning Training Curves

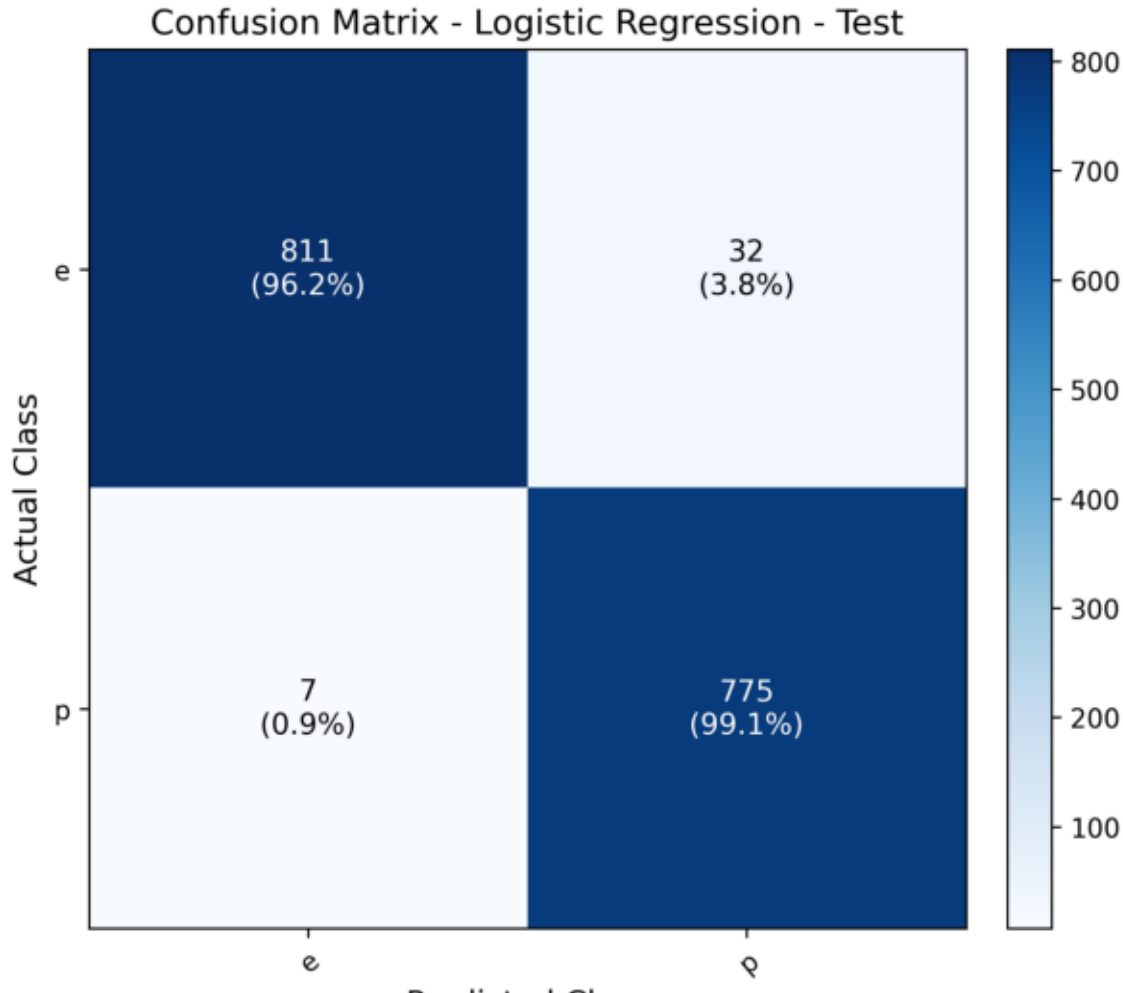




Confusion Matrices - FT-Transformer

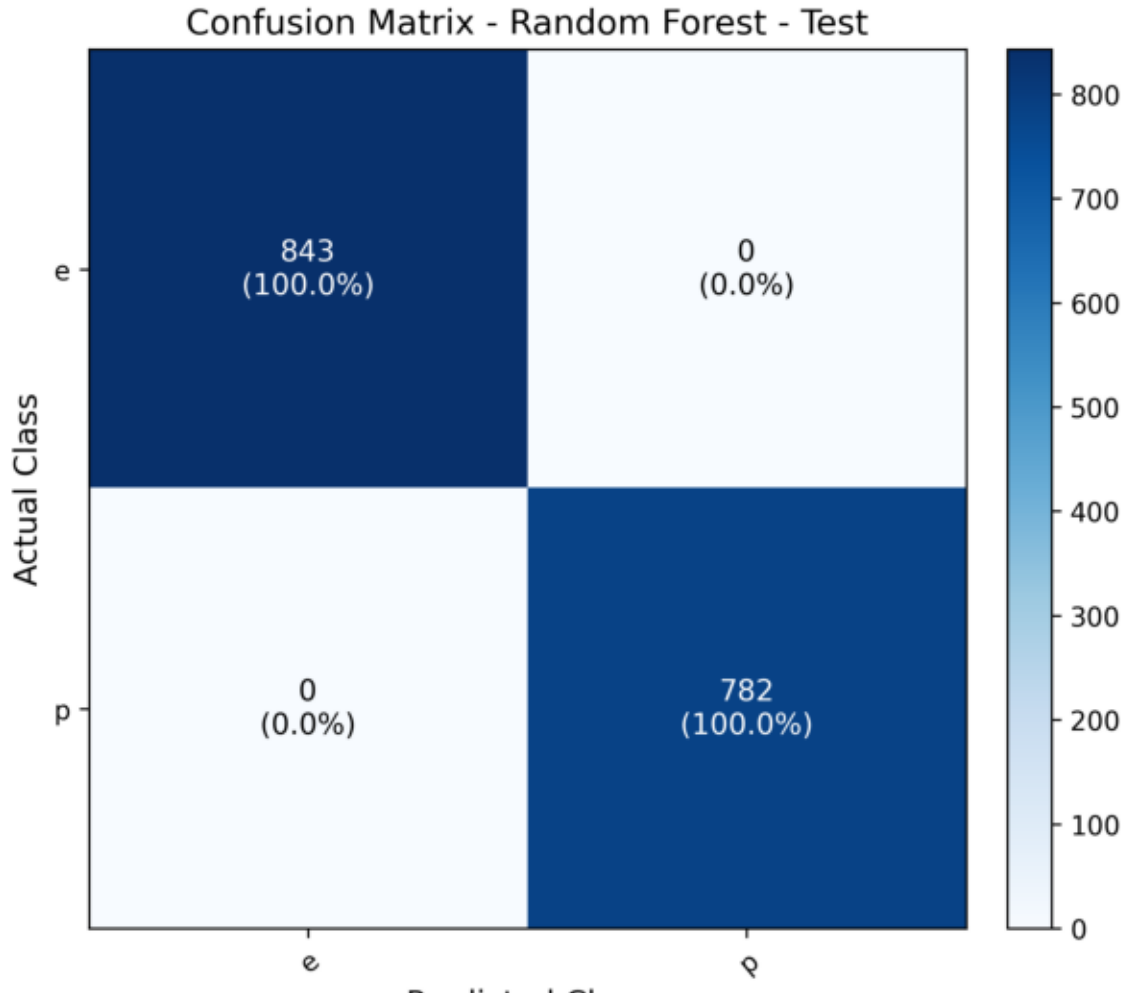


Confusion Matrices - Logistic Regression



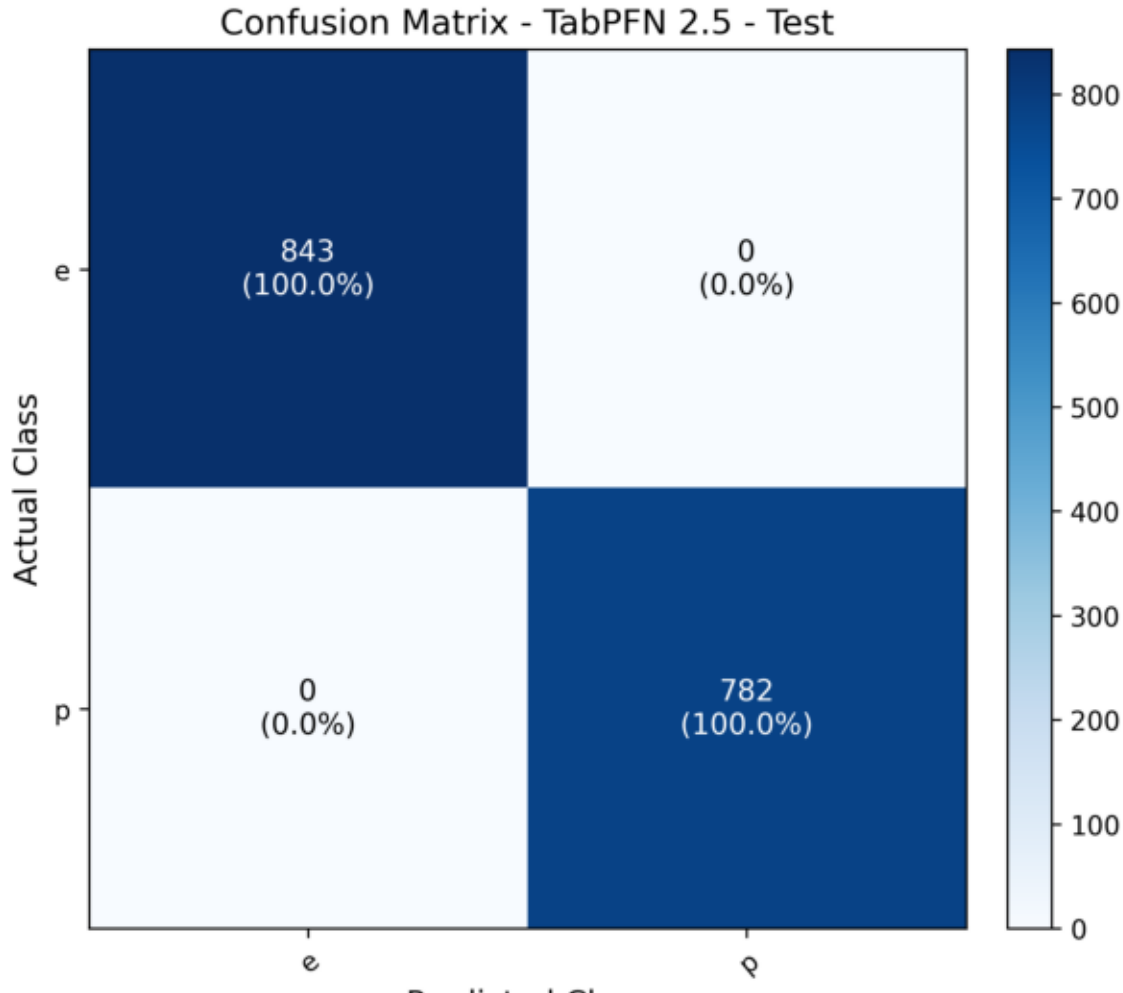
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Confusion Matrices - Random Forest

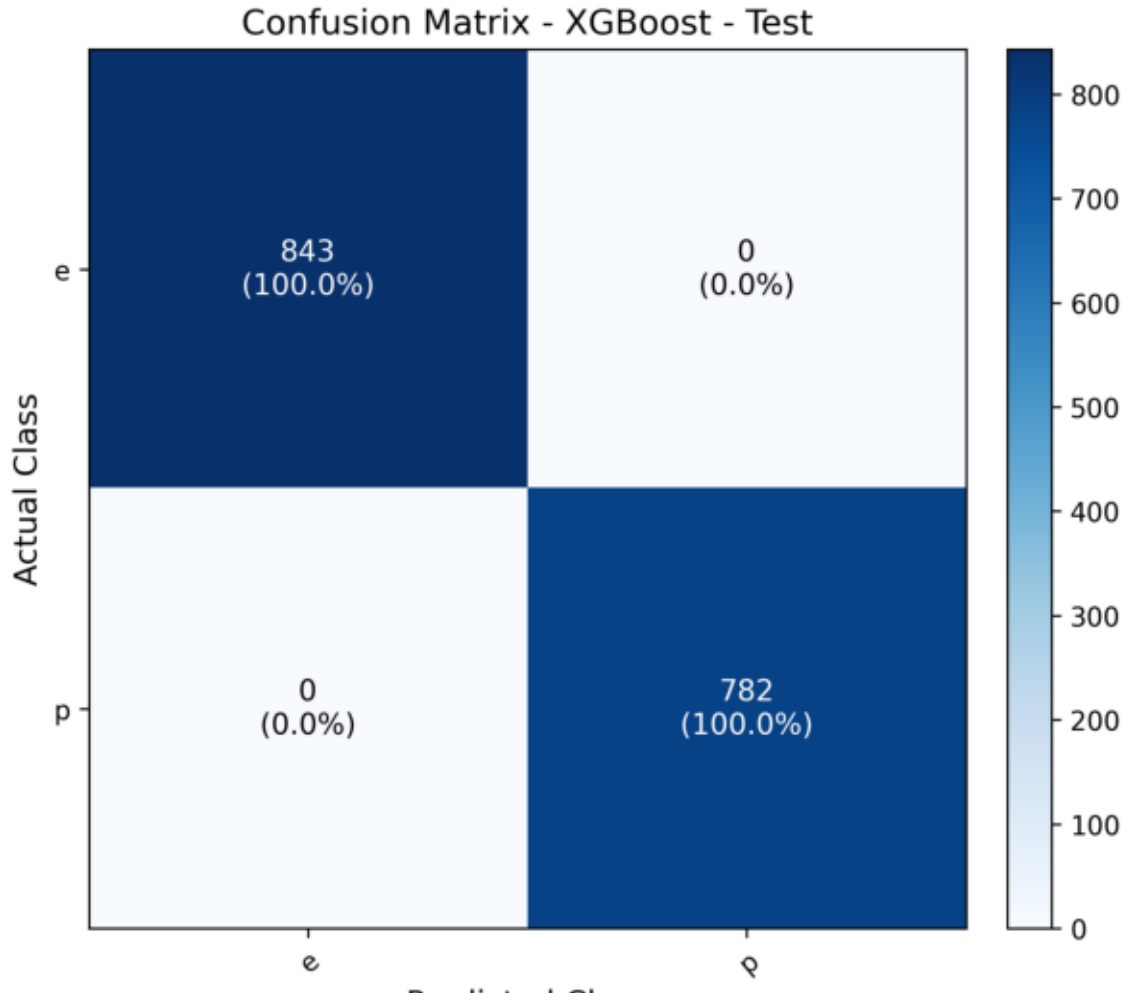


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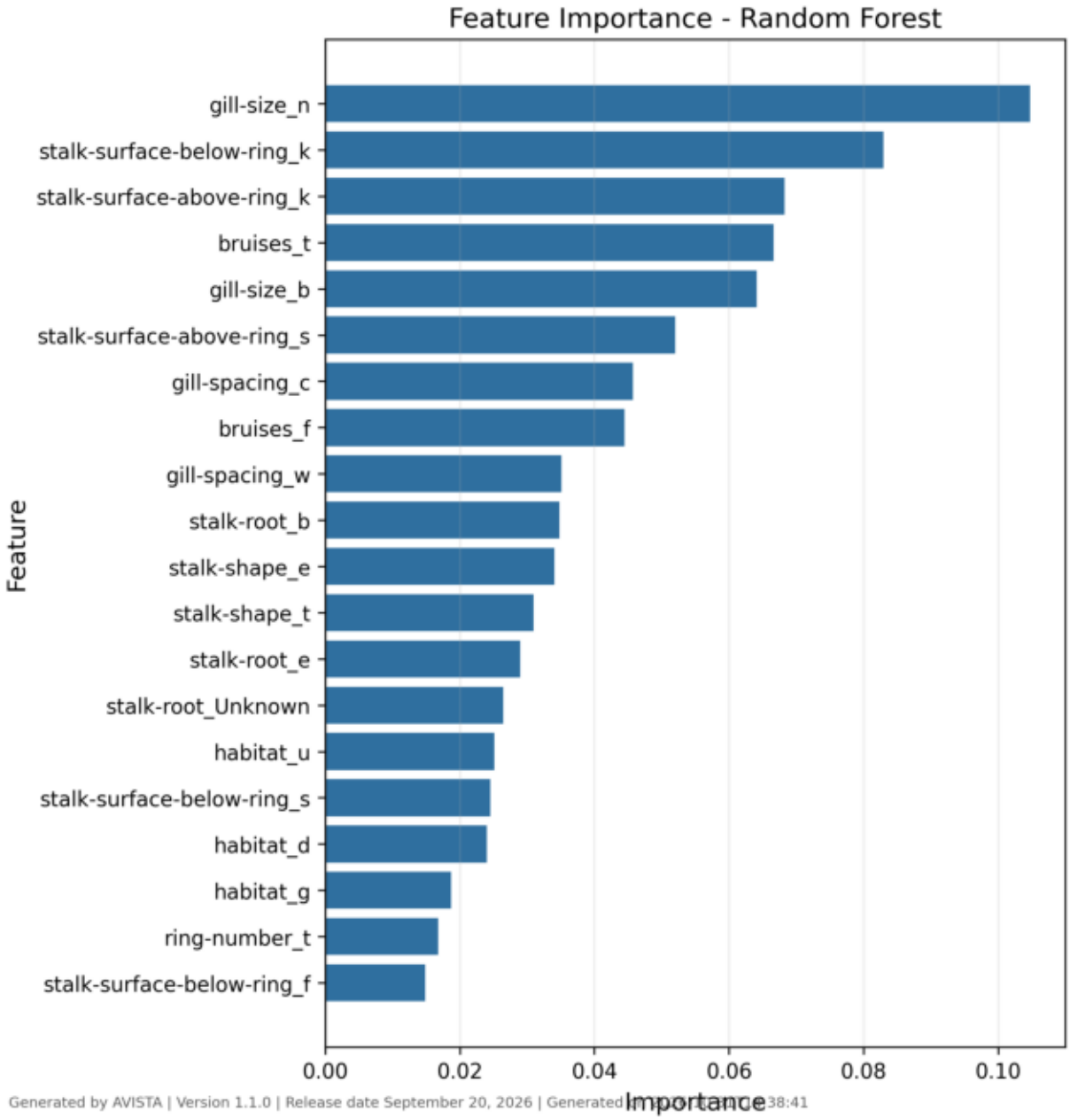
Confusion Matrices - TabPFN 2 5



Confusion Matrices - XGBoost



Feature Importance - Random Forest



Test Classification Report - FT-Transformer

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000

Test Classification Report - Logistic Regression

Class	precision	recall	f1-score	support
e	0.9914	0.9620	0.9765	843.0000
p	0.9603	0.9910	0.9755	782.0000
accuracy	0.9760	0.9760	0.9760	0.9760
macro avg	0.9759	0.9765	0.9760	1625.0000
weighted avg	0.9765	0.9760	0.9760	1625.0000

Test Classification Report - Random Forest

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000

Test Classification Report - TabPFN 2 5

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000

Test Classification Report - XGBoost

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000